Failure Analysis Report

FAIL-0028 — Performance Degradation

1. Failure Summary

Failure ID	FAIL-0028
Failure Type	Performance Degradation
Severity	Major
Component	Firmware Manager
Application	APP-001 — SmartHome
Build Version	6.0.9
Timestamp	2023-02-03 08:48:22
Environment	ENV-004 — XR Lab
OS Version	One UI 6.0
Network	WiFi-6
Resolution	In Progress

2. Test Case Reference

Test Case ID	TC-0257
Title	Verify that playlist shuffle completes successfully when the battery level drops...
Execution ID	EXEC-00160
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0201 — Network Behavior — Offline Mode

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete NFC tap-to-pay within the defined SLA.
Observed Behaviour	The system exhibited performance degradation when the session token has expired.

4. Error Details

Error Message:

```
PerformanceViolation: p95 latency=1356ms exceeds SLA of 582ms.
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.MediaCodec.execute(line 174)
```

Log Reference	LOG-50688

5. Executed Steps at Time of Failure

1	Launch the SmartHome application on the test device.	A success toast message is displayed.

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2	Initiate Bluetooth reconnection by pressing the 'Connect' button.	The performance counter logs a value below the SLA threshold.
3	Assert that database record for entity 'ENT-205' has status='ACTI...	The operation completes without error.
4	Observe the status indicator for up to 30 seconds.	No crash or ANR is observed.
5	Re-connect the headset and confirm auto-recovery.	No crash or ANR is observed.

6. Root Cause Analysis

Buffer overflow in the media codec when processing H.265 streams above 4K resolution.

7. Linked Defect

Defect ID	BUG-0419
Title	Performance Degradation in Firmware Manager: PerformanceViolation: p95 latency=1...
Severity	Major
Priority	Critical
Status	OPEN
Root Cause	Buffer overflow in the media codec when processing H.265 streams above 4K resolu...
Resolution	—
Assigned To	ENG-007

8. Historical Similar Failures

HIST-0027	FAIL-0028	Performance Degradation	5.5.2	2023-08-20	12	Resolved
HIST-0326	FAIL-0327	Performance Degradation	6.3.3	2023-05-24	12	Open
HIST-0502	FAIL-0503	Performance Degradation	4.3.0	2023-01-15	14	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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